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sEMG Signals control
Progress Update

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1. Tether Hand demonstration
2. sEMG control architecture
3. sEMG system demonstration
4. Current issues
5. Future works

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Tether IA Right Hand

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System Architecture & Signal Chain

Bio-Signal
Human Muscle
(sEMG source)

→

Sensing
AD8232 Sensor
(Amplification/analog Filter)

→

Data Acquisition
ESP32 (ADC)
(Digital Sampling)

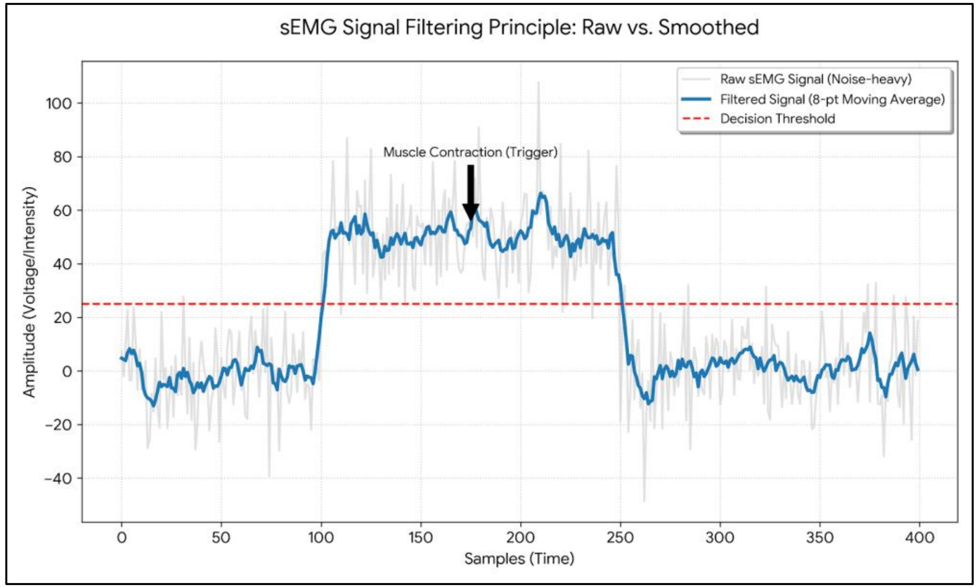
→

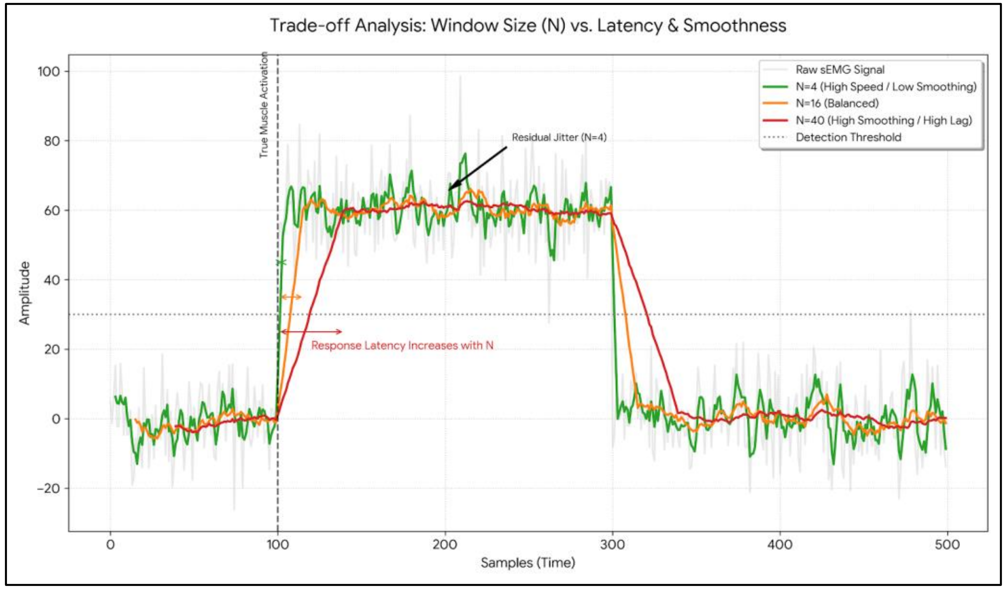
Signal Processing
Python Environment
(Filtering & Logic)

→

Actuation
TetherIA Hand
(Actuation)

Digital Signal Processing



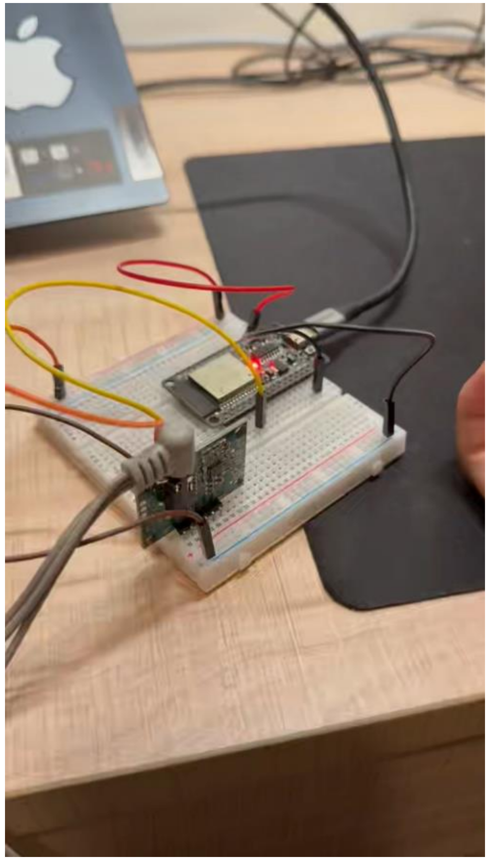


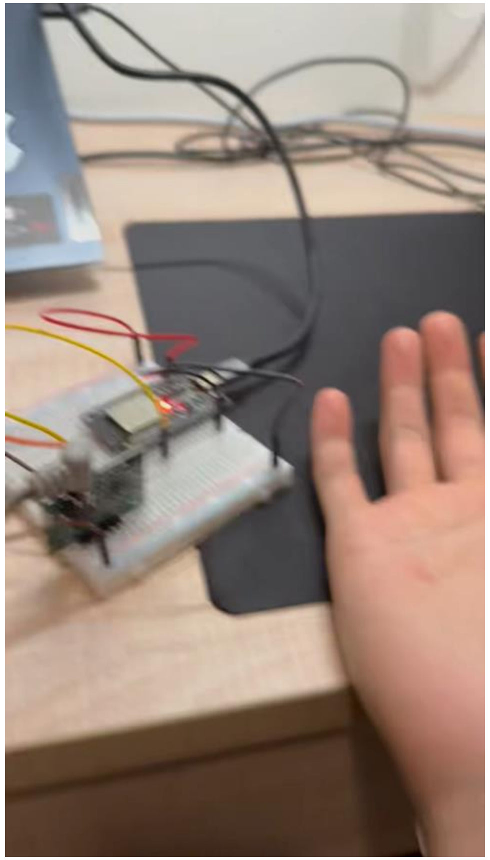
$$y[n] = \frac{1}{N} \sum_{i=0}^{N-1} x[n-i]$$

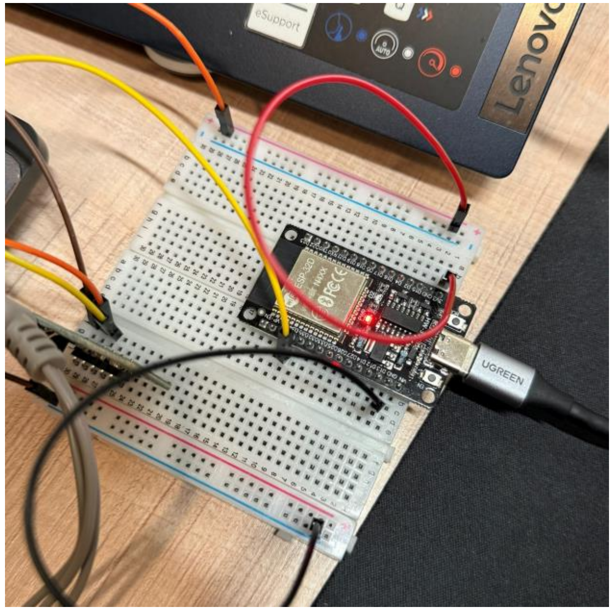
Where *y[n]* is the output average value, *x* is the raw input value, *N* is the length of the Moving Average Window

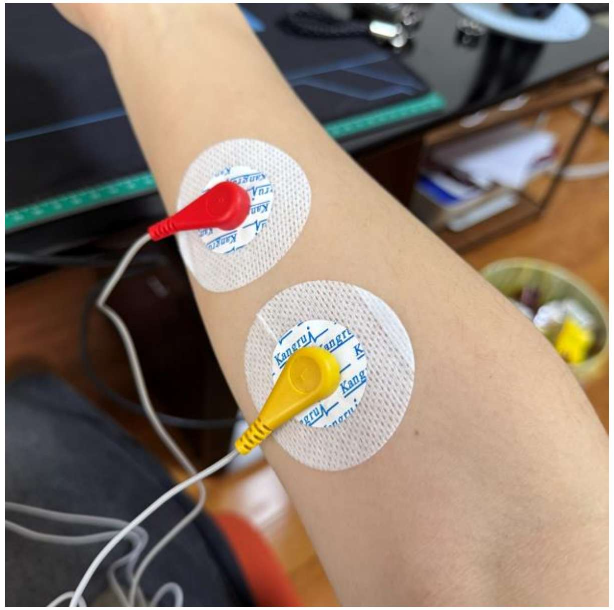
Control Mapping

When averaged signal value > threshold : Giving order of Grasping
When averaged signal value < threshold : Giving order of Opening









Current Issues & Root Cause Analysis

Technical Constraint	Issues	comment
Sensing Hardware	Bandwidth Mismatch	Used low-cost AD8232 (ECG focus) instead of dedicated EMG chips. This leads to a loss of high-frequency components in muscle signals.
Data Acquisition	Single-channel Bottleneck	Unable to resolve co-contraction patterns with one sensor, limiting the 16-DOF hand to simple open/close logic.
Environment	EMI & Signal Drift	Breadboard-based prototype is highly susceptible to 50Hz power-line noise and electromagnetic interference (EMI).
DSP Logic	Latency	The Moving Average filter inherently introduces a response delay proportional to the window size N.
System Loop	Lack of Haptic Feedback	Open-loop position control without torque/force sensing prevents adaptive grasping based on object hardness.

Future Work

- Minimize EMI for Stable Binary Control
- Implement temporal pattern recognition
- Integrated 2x AD8232 Sensing
- RL-driven Robust Grasping